

XRPL Validator — Daily Engineering Report • 2026-07-25

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **6 commits** (xrpl-validator 6)

Engineering changes

- **fix(xrpl-ledger): cap payment delivery at the destination's trust limit** (04a1586 , xrpl-validator) The last TER-class divergence cluster (6: 105740164/105741244/105741859, 2 each) was not the two-hop AMM strand bug the handoff suspected. rippled's final delivery step DirectStepI::maxPaymentFlow (DirectStep.cpp:487) limits what a payment can land on the destination to creditLimit(dst,issuer) minus what it already holds, so a destination at or over its own trust limit receives nothing and the whole flow is ...
- **fix(xrpl-ledger): cross an underfunded maker at book quality, don't rest** (84898d4 , xrpl-validator) The section-1 achieved-quality break (offer.rs) was calibrated for the #105780948 IoC tecKILLED, where the TAKER's own side bounds the fill and the floored output realises worse than the limit. It also fired, wrongly, on a plain offer crossing a maker that is merely UNDERFUNDED at a book quality equal to the taker's limit: the $\leq 1e-7$ round-up on `pay` for the small maker-funds-clamped fill tips the achieved rate ...
- **fix(xrpl-ledger): re-check a maker's actual quality, not just its book page** (7e592e9 , xrpl-validator) rippled compares a crossing maker's ACTUAL `getRate` to the taker's limit, not merely the quantized book-directory level it rests in. An offer can sit in a page that ties the threshold while its exact rate is one ULP worse, and rippled leaves such an offer untouched.
- **fix(xrpl-ledger): honour tfPassive -- don't consume equal-quality makers** (77f6f39 , xrpl-validator) We never read `tfPassive`: a passive offer was crossed like a normal one. A passive offer crosses only STRICTLY better makers and rests behind equal-quality ones (rippled `OfferCreate.cpp:396 ++threshold`). Our rate encoding is inverted (lower = better, monotone in the u64), so tighten the threshold by one ULP for passive offers; the existing `q<=threshold` and `maker-rate` gates then exclude the equal level.
- **fix(xrpl-ledger): implement AMMWithdraw tfWithdrawAll teardown** (a7bd6a0 , xrpl-validator) `tfWithdrawAll` fell through the partial-withdraw path: it withdrew NO assets (no `Amount` fields to loop over) and burned a fixed default `LPToken` amount, leaving the `LPToken` trust line `Modified-at-zero` instead of torn down.
- **fix(differential_probe): hydrate the issuer's owner-dir tail for CheckCash** (7eb0dca , xrpl-validator) `CheckCash` creates the delivered IOU's trust line, which the ISSUER co-owns -- but `native_read_keys` loaded only the casher's owner-dir root, so an issuer with a multi-page directory was never hydrated and `dir_insert` invented a fresh root instead of appending to the existing last page.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Commit-Reveal Batch Auctions for DEX Offers** (dex) Add a tfHidden flag to OfferCreate that stores only sfCommitHash (SHA-256 of taker_pays, taker_gets, account, nonce) instead of plaintext amounts, fil
- **Travel-Rule Payload Requirement for VASP Payments** (compliance) Add an owner-reserved VASPPolicy ledger object (keylet::vaspPolicy(account)), set by a VASPPolicySet transaction, carrying sfTravelRuleThreshold — an
- **Atomic Flash Borrow/Repay for Vault Assets** (lending) Add a VaultFlashBorrow transaction on SingleAssetVault that debits available vault liquidity to the submitting account and records an in-flight sfFlas

Appendix — commits

Time	Repo	Commit	Subject
08:37	xrpl-validator	04a1586	fix(xrpl-ledger): cap payment delivery at the destination's trust limit
10:38	xrpl-validator	84898d4	fix(xrpl-ledger): cross an underfunded maker at book quality, don't rest
13:01	xrpl-validator	7e592e9	fix(xrpl-ledger): re-check a maker's actual quality, not just its book page
16:13	xrpl-validator	77f6f39	fix(xrpl-ledger): honour tfPassive -- don't consume equal-quality makers
18:57	xrpl-validator	a7bd6a0	fix(xrpl-ledger): implement AMMWithdraw tfWithdrawAll teardown
21:02	xrpl-validator	7eb0dca	fix(differential_probe): hydrate the issuer's owner-dir tail for CheckCash